FILMS DATABASE USING MONGODB

Team Members:

23MIS7275-D.RATNA KUMARI

23MIS7242-K.MEENKSHI

COURSE NAME: NOSQL DATABASE

COURSE CODE: CSE3009

SLOT: L39+L40

FACULTY: DR. KARTHIKA NATARAJAN

1.Use Database

```
test> use filmsDB
switched to db filmsDB
```

2.Insert 5 Movies using insertOne()

```
filmsDB> db.films.insertOne({title:"Vikram",year:2022,genre:"Action Thriller",director:"Lokesh Kangaraj",rating:8.3,boxoffice:"BlockBuster",language:"Tamil"})
{
  acknowledged: true,
  insertedId: ObjectId('69766478d9ae35f56b786cd2')
}
```

```
filmsDB> db.films.insertOne({title:"Leo",year:2023,genre:"Action Crime",director:"Lokesh Kangaraj",rating:7.9,boxoffice:"BlockBuster",language:"Tamil"})
{
  acknowledged: true,
  insertedId: ObjectId('6976656ad9ae35f56b786cd3')
}
filmsDB> db.films.insertOne({title:"RRR",year:2022,genre:"Action Drama",director:"S S Rajamouli",rating:8.1,boxoffice:"BlockBuster",language:"Telugu"})
{
  acknowledged: true,
  insertedId: ObjectId('697665c7d9ae35f56b786cd4')
}
filmsDB> db.films.insertOne({title:"3 Idiots",year:2009,genre:"Drama",director:"Rajkumar Hirani",rating:8.4,boxoffice:"BlockBuster",language:"Hindi"})
{
  acknowledged: true,
  insertedId: ObjectId('6976661ed9ae35f56b786cd5')
}
filmsDB> db.films.insertOne({title:"Inception",year:2010,genre:"Science fiction",director:"Chirstopher Nolan",rating:8.8,boxoffice:"BlockBuster",language:"English"})
{
  acknowledged: true,
  insertedId: ObjectId('69766699d9ae35f56b786cd6')
}
```

3.Insert 5 Movies using insertMany()

```
filmsDB>  db.films.insertMany([{title:"Interstellar",year:2014,genre:"Science fiction",director:"Christopher Nolan",rating:8.6,boxoffice:"BlockBuster",language:"English"},{title:"The dark
knight",year:2008,genre:"Action",director:"Christopher Nolan",rating:9.0,boxoffice:"BlockBuster",language:"English"},{title:"Baahubali The Conclusion",year:2017,genre:"Epic",director:"S S
Rajamouli",rating:8.2,boxoffice:"BlockBuster",language:"Telugu"},{title:"Jailer",year:2023,genre:"Action Drama",director:"Nelson DilipKumar",rating:7.8,boxofice:"BlockBuster",language:"Tamil"},
{title:"Enthiran",year:2010,genre:"Science Fiction",director:"Shankar",rating:7.1,boxoffice:"BlockBuster",language:"Tamil"}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69766948d9ae35f56b786cd8'),
    '1': ObjectId('69766948d9ae35f56b786cd9'),
    '2': ObjectId('69766948d9ae35f56b786cda'),
    '3': ObjectId('69766948d9ae35f56b786cdb'),
    '4': ObjectId('69766948d9ae35f56b786cdc')
  }
}
```

4.Display all movie documents

```
filmsDB> db.films.find()
[
  {
    _id: ObjectId('697250bc124509b51e3f8589'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh kanagaraj',
    rating: 8.3,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6972515d124509b51e3f858a'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kanagaraj',
    rating: 7.9,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858e'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxOffice: 'Blockbuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858f'),
    title: 'Inception',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Christopher Nolan',
    rating: 8.8,
    boxOffice: 'Blockbuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766478d9ae35f56b786cd2'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh Kangaraj',
    rating: 8.3,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6976656ad9ae35f56b786cd3'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kangaraj',
    rating: 7.9,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
```

```
{
  _id: ObjectId('697665c7d9ae35f56b786cd4'),
  title: 'RRR',
  year: 2022,
  genre: 'Action Drama',
  director: 'S S Rajamouli',
  rating: 8.1,
  boxoffice: 'BlockBuster',
  language: 'Telugu'
},
{
  _id: ObjectId('6976661ed9ae35f56b786cd5'),
  title: '3 Idiots',
  year: 2009,
  genre: 'Drama',
  director: 'Rajkumar Hirani',
  rating: 8.4,
  boxoffice: 'BlockBuster',
  language: 'Hindi'
},
{
  _id: ObjectId('69766699d9ae35f56b786cd6'),
  title: 'Inception',
  year: 2010,
  genre: 'Science fiction',
  director: 'Chirstopher Nolan',
  rating: 8.8,
  boxoffice: 'BlockBuster',
  language: 'English'
},
{
  _id: ObjectId('697668e3d9ae35f56b786cd7'),
  title: 'RRR',
  year: 2022,
  genre: 'Action Drama',
  director: 'S S Rajamouli',
  rating: 8.1,
  boxoffice: 'BlockBuster',
  language: 'Telugu'
},
{
  _id: ObjectId('69766948d9ae35f56b786cd8'),
  title: 'Interstellar',
  year: 2014,
  genre: 'Science fiction',
  director: 'Christopher Nolan',
  rating: 8.6,
  boxoffice: 'BlockBuster',
  language: 'English'
},
{
  _id: ObjectId('69766948d9ae35f56b786cd9'),
  title: 'The dark knight',
  year: 2008,
  genre: 'Action',
  director: 'Christopher Nolan',
  rating: 9,
  boxoffice: 'BlockBuster',
  language: 'English'
},
```

```
  {
    _id: ObjectId('69766948d9ae35f56b786cda'),
    title: 'Baahubali The Conclusion',
    year: 2017,
    genre: 'Epic',
    director: 'S S Rajamouli',
    rating: 8.2,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdb'),
    title: 'Jailer',
    year: 2023,
    genre: 'Action Drama',
    director: 'Nelson DilipKumar',
    rating: 7.8,
    boxofice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  }
]
```

5.DisplayallmoviesinTamil

```
filmsDB> db.films.find({language:"Tamil"})
[
  {
    _id: ObjectId('697250bc124509b51e3f8589'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh kanagaraj',
    rating: 8.3,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6972515d124509b51e3f858a'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kanagaraj',
    rating: 7.9,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766478d9ae35f56b786cd2'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh Kangaraj',
    rating: 8.3,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6976656ad9ae35f56b786cd3'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kangaraj',
    rating: 7.9,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdb'),
    title: 'Jailer',
    year: 2023,
    genre: 'Action Drama',
    director: 'Nelson DilipKumar',
    rating: 7.8,
    boxofice: 'BlockBuster',
    language: 'Tamil'
  },
```

```
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  }
]
filmsDB> db.films.find({year:{$gt:2015}})
```

6.Display movies released after 2015

```
filmsDB> db.films.find({year:{$gt:2015}})
[
  {
    _id: ObjectId('697250bc124509b51e3f8589'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh kanagaraj',
    rating: 8.3,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6972515d124509b51e3f858a'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kanagaraj',
    rating: 7.9,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766478d9ae35f56b786cd2'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh Kangaraj',
    rating: 8.3,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6976656ad9ae35f56b786cd3'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kangaraj',
    rating: 7.9,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697665c7d9ae35f56b786cd4'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
```

```
{
  _id: ObjectId('697668e3d9ae35f56b786cd7'),
  title: 'RRR',
  year: 2022,
  genre: 'Action Drama',
  director: 'S S Rajamouli',
  rating: 8.1,
  boxoffice: 'BlockBuster',
  language: 'Telugu'
},
{
  _id: ObjectId('69766948d9ae35f56b786cda'),
  title: 'Baahubali The Conclusion',
  year: 2017,
  genre: 'Epic',
  director: 'S S Rajamouli',
  rating: 8.2,
  boxoffice: 'BlockBuster',
  language: 'Telugu'
},
{
  _id: ObjectId('69766948d9ae35f56b786cdb'),
  title: 'Jailer',
  year: 2023,
  genre: 'Action Drama',
  director: 'Nelson DilipKumar',
  rating: 7.8,
  boxofice: 'BlockBuster',
  language: 'Tamil'
}
```

7.Find movies with rating greater than 8

```
filmsDB> db.films.find({rating:{$gt:8}})
[
  {
    _id: ObjectId('697250bc124509b51e3f8589'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh kanagaraj',
    rating: 8.3,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858e'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxOffice: 'Blockbuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858f'),
    title: 'Inception',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Christopher Nolan',
    rating: 8.8,
    boxOffice: 'Blockbuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766478d9ae35f56b786cd2'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh Kangaraj',
    rating: 8.3,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697665c7d9ae35f56b786cd4'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('6976661ed9ae35f56b786cd5'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxoffice: 'BlockBuster',
    language: 'Hindi'
  },
```

```
  {
    _id: ObjectId('69766699d9ae35f56b786cd6'),
    title: 'Inception',
    year: 2010,
    genre: 'Science fiction',
    director: 'Chirstopher Nolan',
    rating: 8.8,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('697668e3d9ae35f56b786cd7'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd8'),
    title: 'Interstellar',
    year: 2014,
    genre: 'Science fiction',
    director: 'Christopher Nolan',
    rating: 8.6,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd9'),
    title: 'The dark knight',
    year: 2008,
    genre: 'Action',
    director: 'Christopher Nolan',
    rating: 9,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cda'),
    title: 'Baahubali The Conclusion',
    year: 2017,
    genre: 'Epic',
    director: 'S S Rajamouli',
    rating: 8.2,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  }
]
```

8.Find movies directed by Christopher Nolan

```
filmsDB> db.films.find({director:"Christopher Nolan"})
[
  {
    _id: ObjectId('697258c6124509b51e3f858f'),
    title: 'Inception',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Christopher Nolan',
    rating: 8.8,
    boxOffice: 'Blockbuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd8'),
    title: 'Interstellar',
    year: 2014,
    genre: 'Science fiction',
    director: 'Christopher Nolan',
    rating: 8.6,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd9'),
    title: 'The dark knight',
    year: 2008,
    genre: 'Action',
    director: 'Christopher Nolan',
    rating: 9,
    boxoffice: 'BlockBuster',
    language: 'English'
  }
]
```

9.Display only title and rating

```
filmsDB> db.films.find({},{title: 1,rating:1,_id:0})
[
  { title: 'Vikram', rating: 8.3 },
  { title: 'Leo', rating: 7.9 },
  { title: '3 Idiots', rating: 8.4 },
  { title: 'Inception', rating: 8.8 },
  { title: 'Vikram', rating: 8.3 },
  { title: 'Leo', rating: 7.9 },
  { title: 'RRR', rating: 8.1 },
  { title: '3 Idiots', rating: 8.4 },
  { title: 'Inception', rating: 8.8 },
  { title: 'RRR', rating: 8.1 },
  { title: 'Interstellar', rating: 8.6 },
  { title: 'The dark knight', rating: 9 },
  { title: 'Baahubali The Conclusion', rating: 8.2 },
  { title: 'Jailer', rating: 7.8 },
  { title: 'Enthiran', rating: 7.1 }
]
```

10.Find Hindi movies with rating above 8

```
filmsDB> db.films.find({language: "Hindi", rating: { $gt: 8 }})
[
  {
    _id: ObjectId('697258c6124509b51e3f858e'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxOffice: 'Blockbuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('6976661ed9ae35f56b786cd5'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxoffice: 'BlockBuster',
    language: 'Hindi'
  }
]
```

11.Find movies released between 2010 and 2020

```
filmsDB> db.films.find({year: { $gte: 2010, $lte: 2020 } }
[
  {
    _id: ObjectId('697258c6124509b51e3f858f'),
    title: 'Inception',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Christopher Nolan',
    rating: 8.8,
    boxOffice: 'Blockbuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766699d9ae35f56b786cd6'),
    title: 'Inception',
    year: 2010,
    genre: 'Science fiction',
    director: 'Chirstopher Nolan',
    rating: 8.8,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd8'),
    title: 'Interstellar',
    year: 2014,
    genre: 'Science fiction',
    director: 'Christopher Nolan',
    rating: 8.6,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cda'),
    title: 'Baahubali The Conclusion',
    year: 2017,
    genre: 'Epic',
    director: 'S S Rajamouli',
    rating: 8.2,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  }
]
```

12.Find movies that are blockbusters

```
filmsDB> db.films.find({boxOffice: "Blockbuster" })
[
  {
    _id: ObjectId('697250bc124509b51e3f8589'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh kanagaraj',
    rating: 8.3,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6972515d124509b51e3f858a'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kanagaraj',
    rating: 7.9,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858e'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxOffice: 'Blockbuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858f'),
    title: 'Inception',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Christopher Nolan',
    rating: 8.8,
    boxOffice: 'Blockbuster',
    language: 'English'
  }
]
```

13.Find movies not in English

```
filmsDB> db.films.find({language: { $ne: "English" } })
[
  {
    _id: ObjectId('697250bc124509b51e3f8589'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh kanagaraj',
    rating: 8.3,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6972515d124509b51e3f858a'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kanagaraj',
    rating: 7.9,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858e'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxOffice: 'Blockbuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('69766478d9ae35f56b786cd2'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh Kangaraj',
    rating: 8.3,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6976656ad9ae35f56b786cd3'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kangaraj',
    rating: 7.9,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697665c7d9ae35f56b786cd4'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
```

```
  {
    _id: ObjectId('6976661ed9ae35f56b786cd5'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxoffice: 'BlockBuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('697668e3d9ae35f56b786cd7'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cda'),
    title: 'Baahubali The Conclusion',
    year: 2017,
    genre: 'Epic',
    director: 'S S Rajamouli',
    rating: 8.2,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdb'),
    title: 'Jailer',
    year: 2023,
    genre: 'Action Drama',
    director: 'Nelson DilipKumar',
    rating: 7.8,
    boxofice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  }
]
```

14.Find movies with rating less than

```
filmsDB> db.films.find({rating: { $lt: 8 } })
[
  {
    _id: ObjectId('6972515d124509b51e3f858a'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kanagaraj',
    rating: 7.9,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6976656ad9ae35f56b786cd3'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kangaraj',
    rating: 7.9,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdb'),
    title: 'Jailer',
    year: 2023,
    genre: 'Action Drama',
    director: 'Nelson DilipKumar',
    rating: 7.8,
    boxofice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  }
]
```

15.Find movies with genre Action

```
filmsDB> db.films.find({ genre: "Action" })
|
[
  {
    _id: ObjectId('69766948d9ae35f56b786cd9'),
    title: 'The dark knight',
    year: 2008,
    genre: 'Action',
    director: 'Christopher Nolan',
    rating: 9,
    boxoffice: 'BlockBuster',
    language: 'English'
  }
]
```

16.Sort movies by rating (ascending)

```
filmsDB> db.films.find().sort({ rating: 1 })
[
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdb'),
    title: 'Jailer',
    year: 2023,
    genre: 'Action Drama',
    director: 'Nelson DilipKumar',
    rating: 7.8,
    boxofice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6972515d124509b51e3f858a'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kanagaraj',
    rating: 7.9,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6976656ad9ae35f56b786cd3'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kangaraj',
    rating: 7.9,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697665c7d9ae35f56b786cd4'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('697668e3d9ae35f56b786cd7'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
```

```
{
  _id: ObjectId('69766948d9ae35f56b786cda'),
  title: 'Baahubali The Conclusion',
  year: 2017,
  genre: 'Epic',
  director: 'S S Rajamouli',
  rating: 8.2,
  boxoffice: 'BlockBuster',
  language: 'Telugu'
},
{
  _id: ObjectId('697250bc124509b51e3f8589'),
  title: 'Vikram',
  year: 2022,
  genre: 'Action Thriller',
  director: 'Lokesh kanagaraj',
  rating: 8.3,
  boxOffice: 'Blockbuster',
  language: 'Tamil'
},
{
  _id: ObjectId('69766478d9ae35f56b786cd2'),
  title: 'Vikram',
  year: 2022,
  genre: 'Action Thriller',
  director: 'Lokesh Kangaraj',
  rating: 8.3,
  boxoffice: 'BlockBuster',
  language: 'Tamil'
},
{
  _id: ObjectId('697258c6124509b51e3f858e'),
  title: '3 Idiots',
  year: 2009,
  genre: 'Drama',
  director: 'Rajkumar Hirani',
  rating: 8.4,
  boxOffice: 'Blockbuster',
  language: 'Hindi'
},
{
  _id: ObjectId('6976661ed9ae35f56b786cd5'),
  title: '3 Idiots',
  year: 2009,
  genre: 'Drama',
  director: 'Rajkumar Hirani',
  rating: 8.4,
  boxoffice: 'BlockBuster',
  language: 'Hindi'
},
{
  _id: ObjectId('69766948d9ae35f56b786cd8'),
  title: 'Interstellar',
  year: 2014,
  genre: 'Science fiction',
  director: 'Christopher Nolan',
  rating: 8.6,
  boxoffice: 'BlockBuster',
  language: 'English'
},
```

```
  {
    _id: ObjectId('697258c6124509b51e3f858f'),
    title: 'Inception',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Christopher Nolan',
    rating: 8.8,
    boxOffice: 'Blockbuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766699d9ae35f56b786cd6'),
    title: 'Inception',
    year: 2010,
    genre: 'Science fiction',
    director: 'Chirstopher Nolan',
    rating: 8.8,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd9'),
    title: 'The dark knight',
    year: 2008,
    genre: 'Action',
    director: 'Christopher Nolan',
    rating: 9,
    boxoffice: 'BlockBuster',
    language: 'English'
  }
]
```

17.Sort movies by year (descending)

```
filmsDB> db.films.find().sort({ year: -1 })
[
  {
    _id: ObjectId('6972515d124509b51e3f858a'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kanagaraj',
    rating: 7.9,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6976656ad9ae35f56b786cd3'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kangaraj',
    rating: 7.9,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdb'),
    title: 'Jailer',
    year: 2023,
    genre: 'Action Drama',
    director: 'Nelson DilipKumar',
    rating: 7.8,
    boxofice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697250bc124509b51e3f8589'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh kanagaraj',
    rating: 8.3,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766478d9ae35f56b786cd2'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh Kangaraj',
    rating: 8.3,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697665c7d9ae35f56b786cd4'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
```

```
  },
  {
    _id: ObjectId('697668e3d9ae35f56b786cd7'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cda'),
    title: 'Baahubali The Conclusion',
    year: 2017,
    genre: 'Epic',
    director: 'S S Rajamouli',
    rating: 8.2,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd8'),
    title: 'Interstellar',
    year: 2014,
    genre: 'Science fiction',
    director: 'Christopher Nolan',
    rating: 8.6,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858f'),
    title: 'Inception',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Christopher Nolan',
    rating: 8.8,
    boxOffice: 'Blockbuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766699d9ae35f56b786cd6'),
    title: 'Inception',
    year: 2010,
    genre: 'Science fiction',
    director: 'Chirstopher Nolan',
    rating: 8.8,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
```

```
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858e'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxOffice: 'Blockbuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('6976661ed9ae35f56b786cd5'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxoffice: 'BlockBuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd9'),
    title: 'The dark knight',
    year: 2008,
    genre: 'Action',
    director: 'Christopher Nolan',
    rating: 9,
    boxoffice: 'BlockBuster',
    language: 'English'
  }
]
```

18.Limit result to 5 movies

```
filmsDB> db.films.find().limit(5)
[
  {
    _id: ObjectId('697250bc124509b51e3f8589'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh kanagaraj',
    rating: 8.3,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6972515d124509b51e3f858a'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kanagaraj',
    rating: 7.9,
    boxOffice: 'Blockbuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858e'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxOffice: 'Blockbuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('697258c6124509b51e3f858f'),
    title: 'Inception',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Christopher Nolan',
    rating: 8.8,
    boxOffice: 'Blockbuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766478d9ae35f56b786cd2'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh Kangaraj',
    rating: 8.3,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  }
]
```

19.Skip first 3 movies

```
filmsDB> db.films.find().skip(3)
[
  {
    _id: ObjectId('697258c6124509b51e3f858f'),
    title: 'Inception',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Christopher Nolan',
    rating: 8.8,
    boxOffice: 'Blockbuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766478d9ae35f56b786cd2'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh Kangaraj',
    rating: 8.3,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('6976656ad9ae35f56b786cd3'),
    title: 'Leo',
    year: 2023,
    genre: 'Action Crime',
    director: 'Lokesh Kangaraj',
    rating: 7.9,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('697665c7d9ae35f56b786cd4'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('6976661ed9ae35f56b786cd5'),
    title: '3 Idiots',
    year: 2009,
    genre: 'Drama',
    director: 'Rajkumar Hirani',
    rating: 8.4,
    boxoffice: 'BlockBuster',
    language: 'Hindi'
  },
  {
    _id: ObjectId('69766699d9ae35f56b786cd6'),
    title: 'Inception',
    year: 2010,
    genre: 'Science fiction',
    director: 'Chirstopher Nolan',
    rating: 8.8,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
```

```
  {
    _id: ObjectId('697668e3d9ae35f56b786cd7'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd8'),
    title: 'Interstellar',
    year: 2014,
    genre: 'Science fiction',
    director: 'Christopher Nolan',
    rating: 8.6,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cd9'),
    title: 'The dark knight',
    year: 2008,
    genre: 'Action',
    director: 'Christopher Nolan',
    rating: 9,
    boxoffice: 'BlockBuster',
    language: 'English'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cda'),
    title: 'Baahubali The Conclusion',
    year: 2017,
    genre: 'Epic',
    director: 'S S Rajamouli',
    rating: 8.2,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdb'),
    title: 'Jailer',
    year: 2023,
    genre: 'Action Drama',
    director: 'Nelson DilipKumar',
    rating: 7.8,
    boxofice: 'BlockBuster',
    language: 'Tamil'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  }
]
```

20.Count total movies

```
filmsDB> db.films.countDocuments()
15
```

21.Count Tamil movie

```
filmsDB> db.films.countDocuments({ language: "Tamil" })
6
```

22.Find movies with rating exactly 8.1

```
filmsDB> db.films.find({ rating: 8.1 })
[
  {
    _id: ObjectId('697665c7d9ae35f56b786cd4'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('697668e3d9ae35f56b786cd7'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  }
]
```

23.Find movies where director name starts with "S"

```
filmsDB> db.films.find({ director: /^S/ })
[
  {
    _id: ObjectId('697665c7d9ae35f56b786cd4'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('697668e3d9ae35f56b786cd7'),
    title: 'RRR',
    year: 2022,
    genre: 'Action Drama',
    director: 'S S Rajamouli',
    rating: 8.1,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cda'),
    title: 'Baahubali The Conclusion',
    year: 2017,
    genre: 'Epic',
    director: 'S S Rajamouli',
    rating: 8.2,
    boxoffice: 'BlockBuster',
    language: 'Telugu'
  },
  {
    _id: ObjectId('69766948d9ae35f56b786cdc'),
    title: 'Enthiran',
    year: 2010,
    genre: 'Science Fiction',
    director: 'Shankar',
    rating: 7.1,
    boxoffice: 'BlockBuster',
    language: 'Tamil'
  }
]
```

24.Update rating of one movie

```
filmsDB> db.films.updateOne({ title: "Leo" }, { $set: { rating: 8.0 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

25.Update boxOffice of one movie

```
filmsDB> db.films.updateOne({ title: "Vikram" }, { $set: { boxOffice: "Super
Blockbuster" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

26.Update language of a movie

```
filmsDB> db.films.updateOne({ title: "RRR" }, { $set: { language: "Telugu/
Tamil" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

27.Update director name

```
filmsDB> db.films.updateOne({ title: "Jailer" }, { $set: { director:
"Nelson" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

28.Increase rating by 0.1

```
filmsDB> db.films.updateOne({ title: "Inception" }, { $inc: { rating:
0.1 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

29.Update genre of one movie

```
filmsDB> db.films.updateOne({ title: "Enthiran" }, { $set: { genre: "Sci-Fi
Action" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

30.Update year of one movie

```
filmsDB> db.films.updateOne({ title: "3 Idiots" }, { $set: { year: 2010 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

31.Update rating for all Tamil movies

```
filmsDB> db.films.updateMany({ language: "Tamil" }, { $set: { rating:
8.0 } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 6,
  modifiedCount: 5,
  upsertedCount: 0
}
```

32.Update boxOffice for all English movies

```
filmsDB> db.films.updateMany({ language: "English" }, { $set: { boxOffice:
"Blockbuster" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 3,
  upsertedCount: 0
}
```

33.Update genre for all action movies

```
filmsDB> db.films.updateMany({ genre: "Action" }, { $set: { genre: "Action
Thriller" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
```

34.Add awards field to one movie

```
filmsDB> db.films.updateOne({ title: "The Dark Knight" }, { $set: { awards:
"Oscar" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

35.Add country field to all movies

```
filmsDB> db.films.updateMany({}, { $set: { country: "India/USA" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 15,
  modifiedCount: 15,
  upsertedCount: 0
}
```

36.Remove awards field from one movie

```
filmsDB> db.films.updateOne({ title: "The Dark Knight" }, { $unset: { awards: "" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
```

37.Rename field boxOffice to collection

```
filmsDB> db.films.updateMany({}, { $rename: { boxOffice: "collection" } })
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 15,
  modifiedCount: 7,
  upsertedCount: 0
}
```

38.Replace entire document

```
filmsDB> db.films.replaceOne(
|   { title: "Leo" },
|   {
|     title: "Leo",
|     year: 2023,
|     genre: "Action",
|     director: "Lokesh",
|     rating: 8.0,
|     collection: "Blockbuster",
|     language: "Tamil"
|   }
| )
|
{
```

39.Delete ONE movie by title (deleteOne)

```
filmsDB> db.films.deleteOne({ title: "Leo" })
{ acknowledged: true, deletedCount: 1 }
filmsDB> db.films.deleteOne({ rating: { $lt: 7 } })
{ acknowledged: true, deletedCount: 0 }
filmsDB> db.films.deleteMany({ language: "Hindi" })
{ acknowledged: true, deletedCount: 2 }
filmsDB> db.films.deleteMany({ year: { $lt: 2005 } })
{ acknowledged: true, deletedCount: 0 }
filmsDB> db.films.deleteMany({ rating: { $lt: 7.5 } })
{ acknowledged: true, deletedCount: 0 }
```

40.Group movies by language

```
filmsDB> db.films.aggregate([{$group:{_id:"4language",count:{$sum:1}}}])
[ { _id: '4language', count: 12 } ]
filmsDB> db.films.aggregate([{ $group: { _id: null, avgRating: { $avg: "$rating" } }}])
[ { _id: null, avgRating: 8.308333333333334 } ]
```

41.Find highest rated movie

```
filmsDB> db.films.find().sort({ rating: -1 }).limit(1)
[
  {
    _id: ObjectId('69766948d9ae35f56b786cd9'),
    title: 'The dark knight',
    year: 2008,
    genre: 'Action Thriller',
    director: 'Christopher Nolan',
    rating: 9,
    boxoffice: 'BlockBuster',
    language: 'English',
    country: 'India/USA',
    collection: 'Blockbuster'
  }
]
```

42.Find lowest rated movie

```
filmsDB> db.films.find().sort({ rating: 1 }).limit(1)
[
  {
    _id: ObjectId('697250bc124509b51e3f8589'),
    title: 'Vikram',
    year: 2022,
    genre: 'Action Thriller',
    director: 'Lokesh kanagaraj',
    rating: 8,
    language: 'Tamil',
    country: 'India/USA',
    collection: 'Super Blockbuster'
  }
]
```

43.Display distinct languages

```
filmsDB> db.films.distinct("language")
[ 'English', 'Tamil', 'Telugu', 'Telugu/Tamil' ]
```